

MATHEMATICAL MODELLING APPROACH TO GIVE INSIGHT TO THE STUDY OF SINGLE VARIANT FOOD-BORNE NIPAH VIRUS DISEASE TRANSMISSION DYNAMICS

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OUTLINE

- Introduction
- Model Formulation
- Discussion and
- Conclusion
- Benefits of this study to Researchers



INTRODUCTION

In the last few years, concerns about newly developing infectious diseases and their possible effects on public health have grown in the scientific and global health communities.

Nipah virus is a disease caused by a member of the Paramyxoviridae family, specifically the Henipavirus genus. The virus was initially discovered in 1999 in the Malaysian settlement of Nipah, which is located on the island of Pulau Pangkor off the west coast of Peninsula Malaysia [1,2].



Research Objectives for the Study

The study on the mathematical modeling of Nipah virus transmission dynamics in Nigeria aims to achieve the following objectives:

- ① **To Develop a Mathematical model for Analyzing the Impact of Nipah Virus in a susceptible community**
- ② **To Estimate Key Epidemiological Parameters for Effective Disease Control and Policy Formulation**
- ③ **To Propose Data-Driven Public Health Interventions for Mitigating the Spread of Nipah Virus in a susceptible population**



Scope and framework of the study

This study focuses on the epidemiology of Nipah virus disease, with a particular emphasis on understanding how the virus is transmitted through contaminated food items. The research investigates:

- Food handling practices and processes that may contribute to contamination.
- Key risk factors and potential intervention points.

To achieve the study's objectives, a deterministic mathematical epidemiological model is formulated using the SIR (Susceptible-Exposed-Infected-Recovered) framework. This framework is based on the SIR compartmental model invented by Kermack and McKendrick in 1927.



Flow Diagram

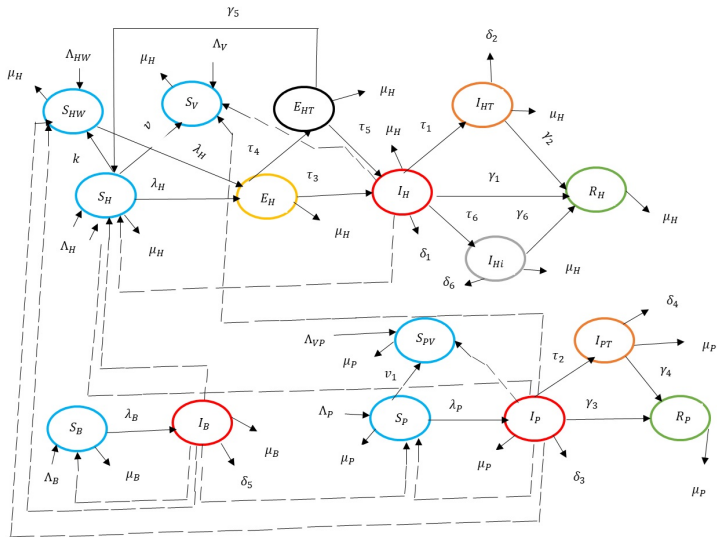


Figure 1: Flow Diagram for Nipah virus transmission dynamics

Selected Assumptions of the Model I

- Those recruited into the system are not infected or contacted the disease.
- We focus on a single strain Nipah virus, like the Nipah virus Malaysia (NiV_M) strain.
- There is a susceptible class for human, susceptible human drinking palm wine, and also susceptible humans vaccinated.
- There is an Intensive supportive care for those infected with the virus that goes for treatment.
- The treatment given to those infected is the combinational therapy method, because as the time of this study there is no single specific approach or method for treating Nipah virus disease.
- We anticipate the development of vaccine for treatment of the disease, which we integrated in our study.
- We hope to use mass media campaign as well as geographical control (on areas with high alert on Nipah virus during an outbreak), as our control for this study.
- Humans can contact the disease by contact with infected bats, infected pigs or infected humans.

NiV Model for the Study

$$\frac{dS_H}{dt} = \Lambda_H + \gamma_5 E_{HT} - \nu S_H - \lambda_H S_H - k S_H - \mu_H S_H \quad (3.1)$$

$$\frac{dS_V}{dt} = \Lambda_V + \nu S_H - \mu_H S_V \quad (3.2)$$

$$\frac{dS_{HW}}{dt} = \Lambda_{HW} + k S_H - \lambda_H S_{HW} - \mu_H S_{HW} \quad (3.3)$$

$$\frac{dE_H}{dt} = \lambda_H S_H + \lambda_H S_{HW} - [\tau_4 + \tau_3 + \mu_H] E_H \quad (3.4)$$

$$\frac{dE_{HT}}{dt} = \tau_4 E_H - [\tau_5 + \gamma_5 + \mu_H] E_{HT} \quad (3.5)$$

$$\frac{dI_H}{dt} = \tau_3 E_H + \tau_5 E_{HT} - [\delta_1 + \tau_1 + \mu_H] I_H \quad (3.6)$$

$$\frac{dI_{HT}}{dt} = \tau_1 I_H - [\gamma_2 + \delta_2 + \tau_6 + \mu_H] I_{HT} \quad (3.7)$$

$$\frac{dI_{Hi}}{dt} = \tau_6 I_{HT} - [\gamma_6 + \delta_6 + \mu_H] I_{Hi} \quad (3.8)$$

$$\frac{dR_H}{dt} = \gamma_2 I_{HT} + \gamma_6 I_{Hi} - \mu_H R_H \quad (3.9)$$

$$\frac{dS_P}{dt} = \Lambda_P - \lambda_P S_P - [\nu_1 + \mu_P] S_P \quad (3.10)$$

NiV Model for the Study Cont'd

$$\frac{dS_{VP}}{dt} = \Lambda_{VP} + v_1 S_P - \mu_P S_{VP} \quad (3.11)$$

$$\frac{dI_P}{dt} = \lambda_P S_P - [\gamma_3 + \delta_3 + \tau_2 + \mu_P] I_P \quad (3.12)$$

$$\frac{dI_{PT}}{dt} = \tau_2 I_P - [\gamma_4 + \delta_4 + \mu_P] I_{PT} \quad (3.13)$$

$$\frac{dR_P}{dt} = \gamma_3 I_P + \gamma_4 I_{PT} - \mu_P R_P \quad (3.14)$$

$$\frac{dS_B}{dt} = \Lambda_B - \lambda_B S_B - \mu_B S_B \quad (3.15)$$

$$\frac{dI_B}{dt} = \lambda_B S_B - [\delta_5 + \mu_B] I_B \quad (3.16)$$

Forces of infection:

$$\lambda_H = \frac{\beta_1 [n_0 I_H + n_1 I_P + n_2 I_B]}{N_H} \quad (3.17)$$

$$\lambda_P = \frac{\beta_2 [n_1 I_P + n_2 I_B]}{N_P} \quad (3.18)$$

$$\lambda_B = \frac{\beta_3 n_2 I_B}{N_B} \quad (3.19)$$

Model Analysis I

After developing the model equations, we inspect the model's well-posedness by demonstrating that the model satisfied the constraints on the positivity of the model's variables as well as the constant remaining of the variables in the region of existence of the variables [3, 4, 5]. As a result, we get the following: Invariant Region of solution Since equations (1 to 16) reflect the general population, it is necessary to emphasize that the associated population sizes can never be negative. Thus, we establish the invariance of Ω as follows:

$$\Omega = \left\{ \begin{array}{l} (S_H, S_V, S_{HW}, E_H, E_{HT}, I_H, I_{HT}, I_{Hi}, R_H, \\ S_P, S_{PV}, I_P, I_{PT}, R_P, S_B, I_B) \in \mathbb{R}_+^{16} \end{array} \right\}$$

then

$$\left\{ \begin{array}{l} (S_H, S_V, S_{HW}, E_H, E_{HT}, I_H, I_{HT}, I_{Hi}, R_H, \\ S_P, S_{PV}, I_P, I_{PT}, R_P, S_B, I_B) \text{ and } N(t) \end{array} \right\} \geq 0, \forall t \geq 0$$

In our proposed model, we deploy the trilogy approach of the human, pig, and bat populations, ensuring that all state variables remain non-negative at all times t .

Theorem

Suppose we have a positive region, Ω , which is defined as:

$$\Omega = \left\{ \begin{array}{l} (S_H, S_V, S_{HW}, E_H, E_{HQ}, I_H, I_{HT}, I_{Hi}, R_H, \\ S_P, S_{PV}, I_P, I_{PT}, R_P, S_B, I_B) \in \mathbb{R}_+^{16} \end{array} \right\} \geq 0 \quad (1)$$

Model Analysis III

Proof.

We show that Ω is positively invariant so that it is sufficient to consider the dynamics of the model system.

From (1):

$$\frac{dS_H}{dt} = \Lambda_H + \gamma_5 E_{HT} - \lambda_H S_H - k S_H - (v + \mu_H) S_H$$

Without loss of generality, we have the following inequality:

This implies that

$$S_H \geq 0 \tag{2}$$

The solution is bounded by zero below and is positive for all $t > 0$. We therefore continue using this same approach to prove for the remaining variables. □

Basic Reproductive number (R_0) for the study

Having determined model meaningfulness, and its equilibrium points, it is necessary to compute the disease basic reproduction number (R_0) and also assess the sensitivity of the model's parameters.

we then have:

$$R_0 = \frac{\beta_1 n_0 ([\Lambda_H \mu_H] + \Lambda_H W[k + \nu + \mu_H] + k \Lambda_H)}{[\tau_4 + \tau_3 + \mu_H][\tau_5 + \gamma_5 + \mu_H][\gamma_1 + \delta_1 + \tau_1 + \mu_H] \mu_H[k + \nu + \mu_H]}. \quad (3)$$



Sensitivity Index Calculation

- In this section, we will apply the normalized forward-sensitivity index: where variable, v , depends on a parameter, p , which is expressed as:

$$r_p^v = \frac{\partial v}{\partial p} \times \frac{p}{v}$$

- Sensitivity indices of R_0 with respect to model parameters.
- Positive sign: This means that an increase in the parameter value leads to an increase in R_0 .
- Negative sign: indicates an increase in the parameter value leads to a decrease in R_0 .

Sensitivity Index Results

Table: Sensitivity index of R_0 with respect to some parameters of the model

Parameters	Numerical Value	Sensitivity Index
μ_H	0.0000421	-0.998201
τ_1	0.0001	-0.0007234
β_1	0.0002	1.0000000
γ_1	0.58	-0.3365
n_0	0.000001	1.0000000

Economic Implications of Parameters

- β_1 (Exposed rate):
 - Health Care Costs: Increased demand for medical care.
 - Workforce Impact: Reduced productivity.
 - Preventive Measures: Investment in public health campaigns.
- n_0 (Modification parameter):
 - Public Health Interventions: Improved sanitation and PPE.
 - Resource Allocation: Fewer cases allow better resource use.
 - Economic Stability: Maintains a productive workforce.

Focus on Specific Parameters Affecting R_0

- n_0 (Modification parameter): Decreasing human infectiousness.
- τ_5 (Rate at which exposed human receiving treatment becomes infectious): Effective and timely treatment.
- β_1 (Exposed rate): Reducing exposure through vaccination and public health campaigns.

Parameter Impact on Nipah Virus Epidemiological Model

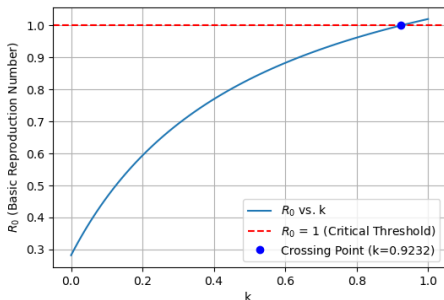


Figure: Graph of reproduction number (R_0) as a function of k .

- At $k = 0.9232$, R_0 crosses the critical threshold of 1.
- Below $k = 0.9232$, $R_0 < 1$: Disease unlikely to spread.
- Above $k = 0.9232$, $R_0 > 1$: Disease may become an epidemic.

Parameter Impact on Nipah Virus Epidemiological Model (Cont.)

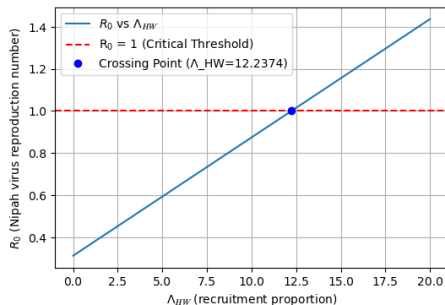


Figure: Plot of reproduction number (R_0) as a function of Λ_{HW} .

Parameter Impact on Nipah Virus Epidemiological Model (Cont.)

Relationship between β_1 and τ_4 with respect to R_0

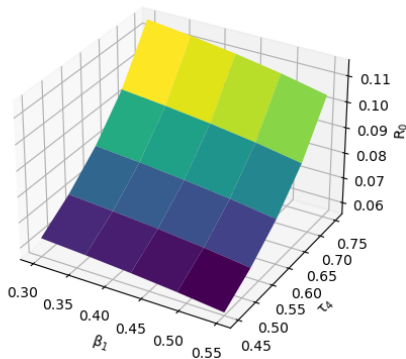


Figure: Simulation of Relationship between β_1 , τ_4 and basic reproduction number (R_0).

Parameter Impact on Nipah Virus Epidemiological Model (Cont.)

Relationship between λ_{HW} and γ_5 with respect to R_0

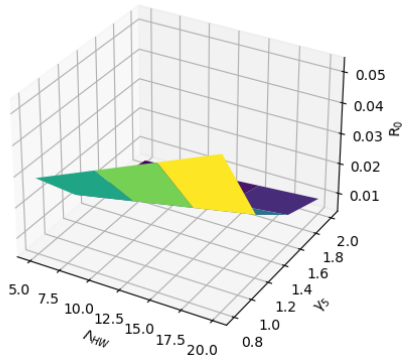


Figure: Relationship between λ_{HW} , γ_5 and reproduction number (R_0).

Conclusion

- By studying the epidemiology of Nipah virus, we developed an accurate mathematical model that captures the majority of the disease traits observed during an epidemic. This model explains the rapid and widespread spread of the virus within endemic populations.
- Two critical factors in reducing the spread of Nipah virus are:
 - The rate of likelihood of consuming contaminated NiV food items (k).
 - The recruitment proportion of susceptible individuals likely to consume tainted food items (Λ_{HW}).
- Through the analysis of equations 1 to 16, we identified threshold values for k and Λ_{HW} . These thresholds provide actionable insights for stakeholders to design effective controls for treatment and transmission of NiV.
- In future publications, we plan to solve this model using various numerical methods, which will further validate the proposed model and enhance its applicability.

- **For Researchers Traveling for Conferences, workshops or Fieldwork:**

- The findings offer a comprehensive framework for understanding the transmission dynamics of the Nipah virus in endemic regions, assisting researchers in effectively designing and conducting their research activities.
 - The identified thresholds for k and Λ_{HW} can guide researchers in assessing the risk involved in handling NiV tainted food items.
 - The model serves as a tool for predicting outbreak scenarios, helping researchers prioritize interventions and allocate resources effectively during fieldwork.
 - By validating the model with numerical methods, researchers can use it to simulate and test the impact of various control measures in real-world settings.
- These insights not only advance scientific understanding but also enhance the safety and effectiveness of research activities in regions affected by Nipah virus.

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